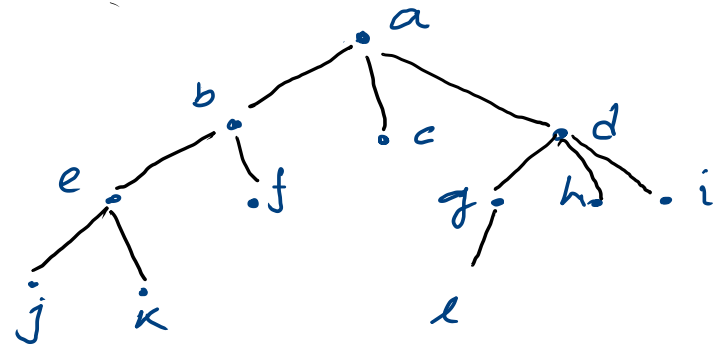


Chapter 11.3 Tree traversal

pre-order: **Root**, $L \rightarrow R$

in-order: **left most**, **Root**, $L \rightarrow R$

post order: $L \rightarrow R$, **Root**



Pre-order

a (b) c (d)
→ a b (e) f c d (g) h i
→ a b e j k f c d g l h i

In-order

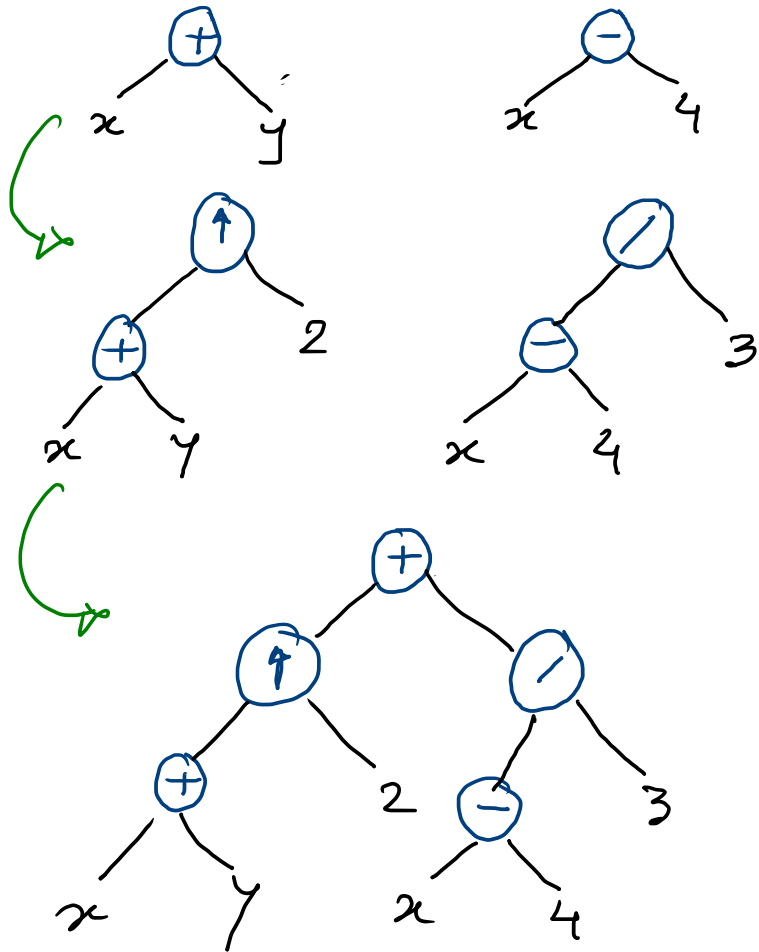
(b) a c (d)
→ (e) b f a c (g) d h i
→ j e k b f a e l g d h i

Post-order

(b) c (d) a
→ (e) f b c (g) h i d a
→ j k e f b c l g h i d a

⑧ Ordered rooted tree for expression

$$\rightarrow ((x+y)^2) + ((x-4)/3) =$$



- prefix (pre-order)

+ ^ /
 $\rightarrow + ^ + 2 / - 3$
 $\rightarrow + ^ + x y 2 / - x 4 3$

- infix (in-order)

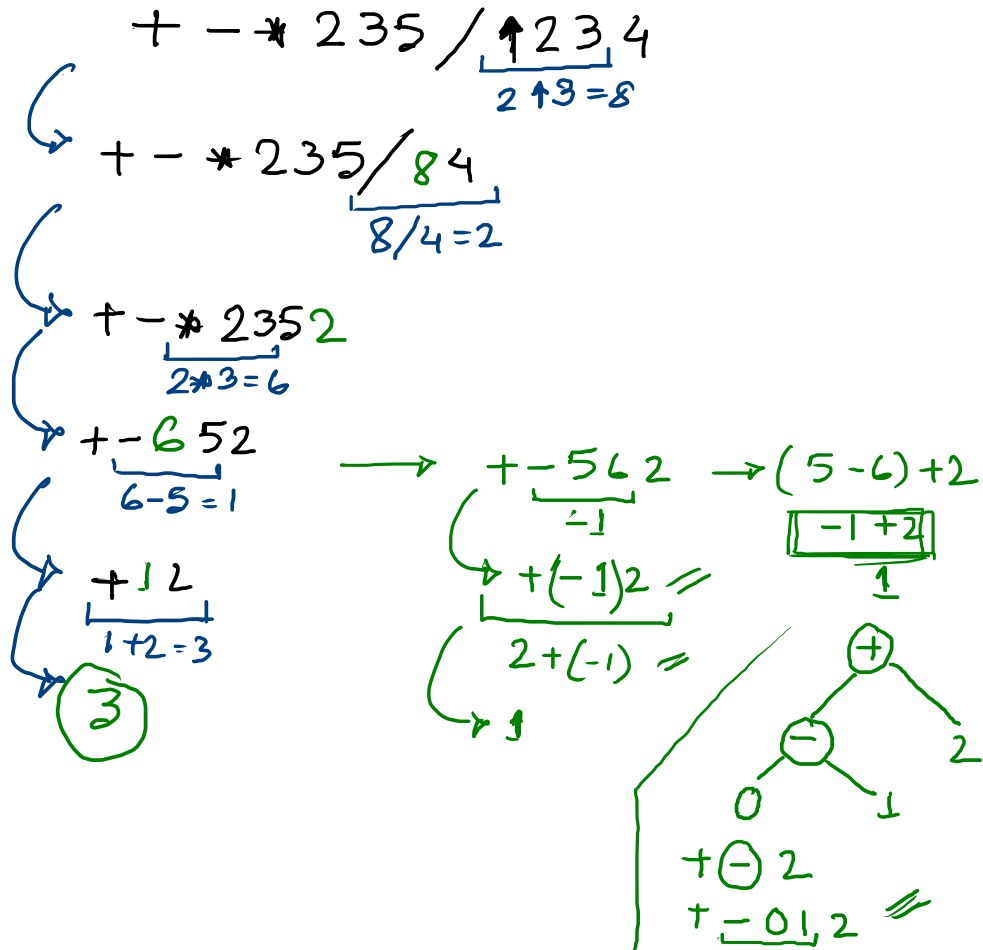
+ ^ /
 $\rightarrow + ^ 2 + - / 3$
 $\rightarrow x + y ^ 2 + x - 4 / 3$

- postfix (post-order)

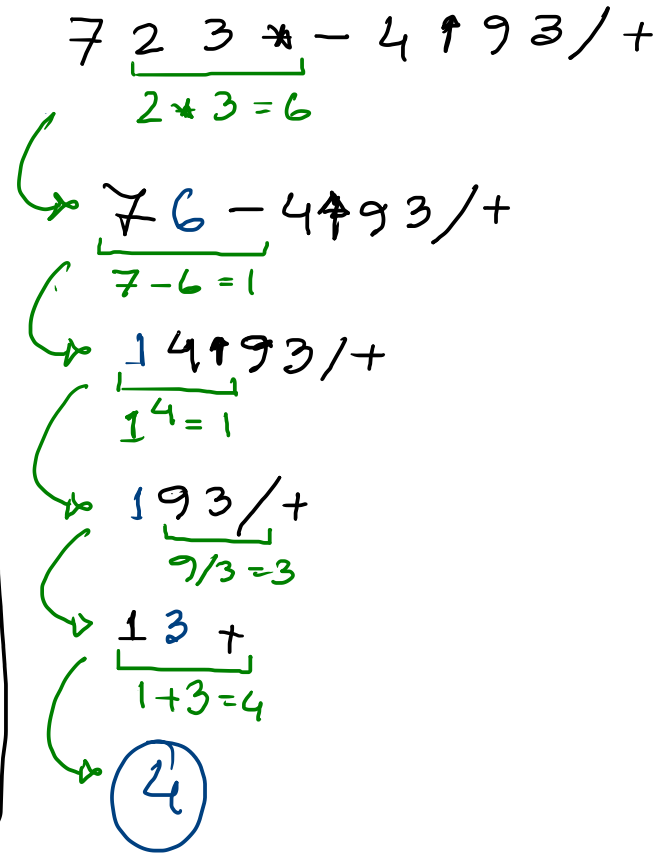
+ ^ / +
 $\rightarrow + 2 ^ - 3 / +$
 $\rightarrow x y + 2 ^ x 4 - 3 / +$

④ Evaluating postfix, prefix

④ Prefix: $R \rightarrow L$



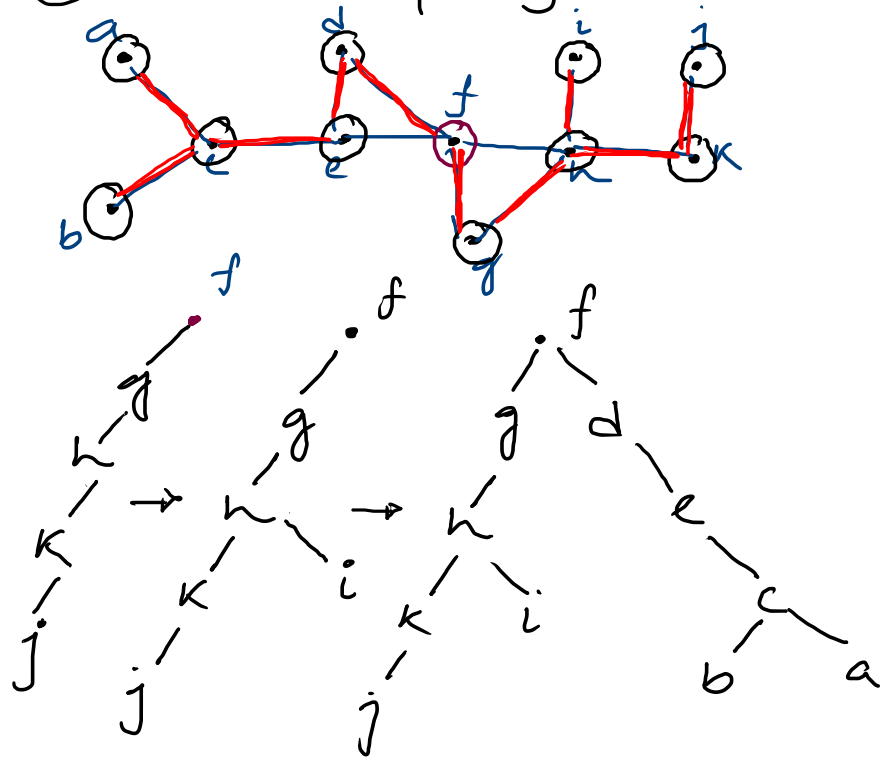
④ Postfix: $L \rightarrow R$



Chapter- 11.4 Spanning tree.

① Spanning Tree: the sub-graph of a graph G containing all the vertices of G .

② DFS $\rightarrow$ Depth first Search



③ BFS - Breadth first Search.

